CALCULUS · UNIT 7: APPLICATIONS OF INTEGRALS

# Glass Vase Studio

*Use area between curves and volumes of revolution to design glass vases and bottles.*

**CP1**

2–3 class periods

Name: _____ Date: _____ Period: _____

## The Scenario

Lumen Glass Studio makes vases, bottles, and paperweights. Each design is a curve spun around an axis, the logo is a region between two curves, and glass is priced by volume.

You will use integrals to calculate the area and volume of each design and price the products.




**Glass Price**

$0.08 per cm³

**Vase Profile**

$r = 4 + \sin(\pi x/10)$

$x$ in cm, $0 \le x \le 20$

## Learning Goals

- I can find areas between curves.
- I can find volumes with the disc and washer methods.
- I can find volumes with known cross sections.

## Project Timeline

| When | What you will do |
|---|---|
| Day 1 | Logo & Labels |
| Day 2 | Vases |
| Day 3 | Paperweight & Production |
| Final | Product Design Portfolio + reflection |

## Part 1: Logo & Labels

**Day 1 ·** Use the Design Specification Cards.

**Design Specification Cards (units: cm)**

| Logo | Bud Vase | Tall Vase |
|---|---|---|
| Between $y = \sqrt{x}$ | $y = \sqrt{x}$ rotated about | $r = 4 + \sin(\pi x/10)$ |
| and $y = x/2$ | the x-axis, $0 \le x \le 16$ | $0 \le x \le 20$ |

| Hollow Vase | Paperweight | Production |
|---|---|---|
| Outer $r = 4 + \sin(\pi x/10)$ | Base: $y = 4 - x^2$, $y \ge 0$ | Rate $p(t) = 6 + 2t$ |
| inner $r = 3.5 + \sin(\pi x/10)$ | square cross sections | vases per hour |

1. Find the area of the logo between $y = \sqrt{x}$ and $y = \frac{x}{2}$.

2. Find the area of the same region by integrating with respect to y.

3. A label is the region between $y = 4 - x^2$ and $y = x + 2$. Find its area.

4. Explain when integrating in y is easier than in x.

## Part 2: Vases

**Day 2 ·** Use the Design Specification Cards.

**5.** Find the volume of the solid tall vase $r = 4 + \sin(\frac{\pi x}{10})$, $0 \le x \le 20$.

**6.** Find the volume of glass in the hollow vase (washer method).

**7.** Find the glass cost of the hollow vase and the volume of water it holds.

**8.** Find the volume of the bud vase $y = \sqrt{x}$ rotated about the x-axis for $0 \le x \le 16$.

## Part 3: Paperweight & Production

**Day 3 ·** Base: region under $y = 4 - x^2$, above $y = 0$. Square cross sections perpendicular to the x-axis.

**9.** Find the volume of the paperweight.

**10.** Find the glass cost.

**11.** Production rate $p(t) = 6 + 2t$ for $0 \le t \le 4$. Find total vases and price the day's glass if each is a hollow vase.

## Final Product: Product Design Portfolio

**Your final product must include:**

- Logo and label areas (both methods)
- Solid and hollow vase volumes
- Paperweight cross-section volume
- Production and cost
- Design portfolio

## Reflection

**R1.** How is the washer method related to the disc method?

**R2.** How do cross sections generalize volume?

## Scoring Rubric

| Criteria | 4 – Exceeds | 3 – Meets | 2 – Approaching | 1 – Beginning |
|---|---|---|---|---|
| **Mathematical Accuracy** | All calculations and solutions are correct; answers include units and are checked. | Nearly all work is correct; minor errors do not change conclusions. | Several errors; some conclusions are affected. | Many errors or missing work. |
| **Reasoning & Justification** | Clear, complete explanations connect every answer to the situation; uses precise vocabulary. | Explanations are mostly clear and connected to the situation. | Explanations are brief or partly unclear. | Little or no explanation. |
| **Application & Design** | Areas and volumes are set up and evaluated correctly and used to price and design products. Goes beyond the minimum with a creative, realistic choice. | Areas and volumes are set up and evaluated correctly and used to price and design products. | Design is incomplete or only partly uses the math from the unit. | Design is missing or unrelated to the unit math. |
| **Communication & Presentation** | Portfolio includes diagrams, integrals, results, and prices. Polished, organized, and visually engaging. | Portfolio includes diagrams, integrals, results, and prices. Organized and easy to follow. | Product is missing required parts or is hard to follow. | Product is mostly missing. |